

PLANITT — Crypto Bot

ML Model Backtesting Report

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Executive Summary

This report backtests the **Crypto Bot** ML pipeline on **10 cryptocurrencies** using 2y of daily OHLCV data. Three ML systems are evaluated:

- 1. Market Regime Detector** — ADX/ATR/Bollinger/EMA-based classifier (Trending Up/Down, Ranging, Volatile)
- 2. Multi-Strategy Consensus Engine** — 9 strategies with weighted voting (RSI, MACD, EMA Trend, Supertrend, Bollinger Squeeze, Ichimoku, Volume Breakout, Stochastic RSI, Confluence)
- 3. DistilBERT Transformer** — Fine-tuned for crypto sentiment classification (Bearish/Neutral/Bullish)

Strategy Weights: Confluence (0.28), Supertrend (0.16), Ichimoku (0.16), EMA Trend (0.13), Volume Breakout (0.12), Bollinger Squeeze (0.08), Stochastic RSI (0.07), MACD (0.05), RSI (0.04)

Aggregate Performance Metrics

Metric	Value	Metric	Value
Regime Accuracy	0.2708	Regime F1	0.2699
Signal Accuracy	0.5153	Signal F1	0.4613
Signal Precision	0.5410	Signal Recall	0.5153
Avg Win Rate	70.5%	Total Trades	58
Avg Return	12.23%	Avg Max Drawdown	-5.14%
Avg Sharpe	1.41	Symbols Tested	10

Per-Symbol Backtesting Results

Symbol	Reg.Acc	Reg.F1	Sig.Acc	Sig.F1	Win%	Trades	Return%	Sharpe	MaxDD%
BTC	0.233	0.286	0.512	0.478	71.4	7	-4.3	-0.92	-8.8
ETH	0.308	0.314	0.479	0.444	60.0	10	-0.1	0.14	-14.6
BNB	0.219	0.266	0.425	0.374	33.3	6	-9.2	-1.70	-12.0
SOL	0.343	0.351	0.488	0.423	80.0	5	25.8	2.69	-0.9
XRP	0.192	0.187	0.585	0.521	75.0	4	20.4	1.45	-6.5
ADA	0.336	0.326	0.722	0.681	100.0	4	28.5	2.71	0.0
DOGE	0.281	0.291	0.441	0.395	100.0	7	29.7	2.93	0.0
AVAX	0.322	0.321	0.486	0.410	60.0	5	15.0	1.61	-6.4
DOT	0.253	0.238	0.600	0.543	50.0	6	10.2	1.55	-1.5
MATIC	0.222	0.117	0.414	0.343	75.0	4	6.3	3.60	-0.7

Methodology & Technical Notes

Data Source: Yahoo Finance — 2 years daily OHLCV for top 10 cryptocurrencies.

Train/Test: 80/20 chronological split (no look-ahead bias).

Regime Detection: ADX-14 for trend strength, ATR-14 for volatility, Bollinger Bandwidth for squeeze, EMA-50 for trend direction. 4 regimes: Trending Up, Trending Down, Ranging, Volatile.

Strategies: 9 independent strategies — RSI (oversold/overbought), MACD (histogram crossover), EMA Trend (21/50 cross), Supertrend (ATR-based trailing stop), Bollinger Squeeze (bandwidth contraction), Stochastic RSI, Volume Breakout (1.5x avg volume), Ichimoku Cloud, Confluence (3+ strategy agreement).

Consensus: Weighted voting across strategies. Signal emitted when weighted score > 0.15.

DistilBERT Sentiment: Fine-tuned on crypto news for 3-class classification. Evaluated via expanded lexicon proxy in backtest.

Trading Sim: Long-only. Enter on BUY consensus, exit on SELL/HOLD. Initial \$100,000. No leverage.

Disclaimer: Past performance does not guarantee future results. Crypto markets are highly volatile. This report uses historical data for informational purposes only and does not constitute investment advice.